I. Question (Algorithm Analysis)

Given the following function

```
int algorithm1(int N){
    if (N == 0)
        return 0;
    int sum = 0;
    for (int i = 0; i < N; i++)
        sum++;
    for (int i = 0; i < N; i++)
        for (int j = 0; j < N; j++)
            sum++;
    for (int i = 0; i < 9; i++)
        sum += algorithm1(N / 3);
    return sum;
}
```

What is the time complexity of algorithm1? Explain your result. Use the master theorem.

II. Question (Linked List)

Write the algorithm Sieve of Eratosthenes to extract prime numbers using singly linked list. The algorithm works as follows:

- The user enters a number N.
- Put all numbers starting from 2 to N in a linked list.
- While the linked list contains numbers
  - Remove the first element p from the linked list. Print it (It is prime).
  - Remove all elements from the linked list which are divisible by p. Do not print them.

```
void eratosthenes()
```

III. Question (Trees)

Write a recursive method in TreeNode class that finds the number of duplicate keys in a binary search tree. Assume that if a key is duplicate, it occurs at most twice. Hint: The duplicate of a key is either the maximum number on its left subtree or the minimum number on its right subtree.

```
int numberOfDuplicates()
```

IV. Question (Graphs)

Write a method which checks if two graphs are the same. Assume that, the method is written in the Adjacency list representation of a graph.

```
boolean isSame(Graph g)
```

V. Question (Sorting)

Suppose arrays representing sets A and B are both sorted. Write a linear time method that finds the minimum difference between any element from A and any element from B.

```
int minDifference(int[] A, int[] B)
```

VI. Question (Sorting)

Write a method which returns the sorted form of the linked list (as a new linked list), which contains only numbers 1, 2, and 3. Your algorithm should run in linear time $\mathcal{O}(N)$.

```
LinkedList sortLinkedList(LinkedList list)
```